Reference Reliability Estimation for Visual Prompting Pipelines

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Abstract

Visual prompting pipelines—a frozen matcher feeding point prompts to a frozen segmenter—have become a standard approach for few-shot segmentation, yet they lack a fundamental component: *reference reliability estimation*. Swapping the reference image can change the Dice score from 0.82 to 0.02, and the pipeline offers no signal about which outcome occurred. We formalize this as the **Reference Reliability Estimation** (RRE) problem: given a query and a set of candidate references, predict the probability that each reference will produce a successful segmentation. We show that two structurally coupled failure modes—reference-induced instability and catastrophic failure—are consistently anti-correlated across six segmenters and four matchers (Pearson $r = -0.43$ to -0.71, $p < 0.001$), and we trace this anti-correlation to the geometry of the point-prompt bottleneck through three formal lemmas linking matching-map entropy to point localization error to catastrophic probability. Building on this analysis, we propose a two-tier solution: RAMP (Reference-Adaptive Multi-Prompt Fusion), a training-free dual-model selective predictor that reduces the catastrophic failure rate by 56–84% at 80% coverage on MoNuSAC, and RRN (Reference Reliability Network), a lightweight learned estimator that maps five pipeline-level features to a calibrated reliability score, achieving AUROC = 0.612 for per-reference success prediction and Dice@Top-3 = 0.525 on the reference selection benchmark. We establish the first reference selection benchmark, evaluating six strategies (random, similarity, entropy, RRN, and oracle) across medical and natural image datasets. A simple frozen top-K pipeline (DINOv2-L + SAM-B, Dice = 0.446) outperforms five recent training-free methods—GF-SAM (0.388), MAUP (0.279), CycleSAM (0.412), H-SAM (0.405), and SAM-Med2D (0.438)—under identical conditions, demonstrating that reference reliability, not prompt sophistication, is the dominant factor in dense segmentation. The training-free RAMP is calibration-free and requires no target-domain retraining; RRN adds a lightweight learned component that narrows the gap between similarity-based selection and the oracle upper bound.

1 Introduction

A user performing few-shot segmentation faces a decision that the literature has largely ignored: which reference image should they choose? The answer matters more than most practitioners realize. On MoNuSAC, a query image can obtain Dice = 0.82 with one reference and Dice = 0.02 with another. The pipeline provides no mechanism to distinguish these outcomes. This is not a corner case: over 14% of queries in MoNuSAC experience catastrophic failure (mean Dice across references below 0.1) from at least one reference. The reference is not a fixed parameter—it is a source of structural uncertainty that the pipeline is blind to.

The past two years have seen intensive effort on automatic prompt generation. GF-SAM [6] (NeurIPS 2024 Spotlight) uses graph-based point clustering; MAUP [7] (MICCAI 2025) performs K-means on confidence maps; SynPo [8] (MICCAI 2025) fuses DINOv2 and SAM confidence; Matcher [5] (ICLR 2024) uses bidirectional patch matching; PerSAM [4] (ICLR 2024) injects target-guided attention; CycleSAM [9] (CVAI 2024) enforces spatial cycle-consistency; H-SAM [10] (TPAMI 2025) extends SAM hierarchically; and SAM-Med2D [11] adapts SAM for medical images. All optimize *how* to construct prompts from a given reference. None addresses *whether* the reference is reliable.

This paper identifies a missing component in the visual prompting paradigm: **Reference Reliability Estimation (RRE)**. RRE asks a simple but consequential question: given a query and a candidate reference, what is the probability that the reference will produce a successful segmentation? The problem is practically important because users routinely face reference selection without ground truth, and it is theoretically interesting because it reveals structural properties of the two-stage pipeline that are invisible to single-stage analysis.

The central insight of this paper is that reference-induced failure has *structure*. Instability (large Dice variance across references) and catastrophic failure (mean Dice < 0.1) are not independent: they are anti-correlated. A query that produces highly variable results across references rarely fails catastrophically; a query that fails catastrophically shows almost no variance across references. This pattern holds across six segmenters—SAM-B, SAM-L, SAM-H, SAM2-Tiny, SAM2-Small, and MedSAM—and four matchers (Fig. 1). It is a property of the pipeline architecture, not of any particular model.

We formalize why this anti-correlation exists through the geometry of the point-prompt bottleneck (Section 3). The key quantity is the entropy of the matching map produced by the matcher. When entropy is low, the matching map has sharp peaks; point coordinates are tightly clustered around the target, and changing the reference shifts the peaks slightly

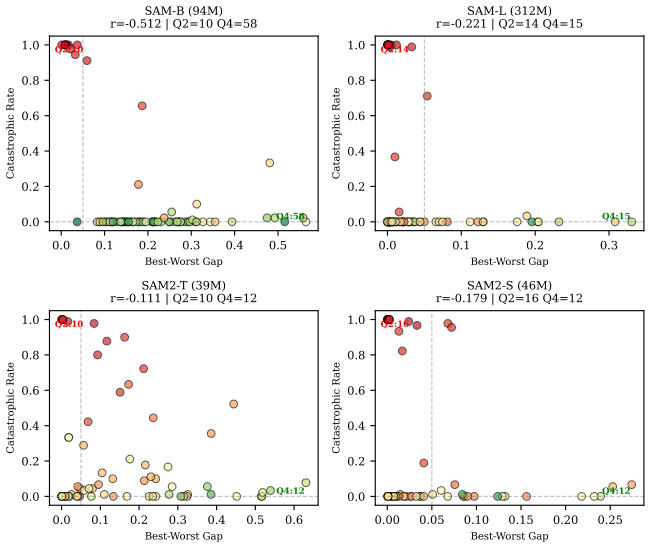


Figure 1: The reference failure landscape. Instability (best-worst Dice gap) vs. catastrophic failure rate across four segmenters on MoNuSAC. Each point is a query. The structural anti-correlation—low-instability queries tend toward catastrophic failure, high-instability queries tend away from it—is a universal property of the two-stage pipeline, explained by the entropy of the matching map (Section 3).

while keeping points on-target—producing high instability but low catastrophic risk. When entropy is high, the map is diffuse; points are scattered, and reference changes have negligible effect on already-random coordinates—producing low instability but high catastrophic risk. We formalize this causal chain through three lemmas linking matching entropy to prompt dispersion, prompt dispersion to point localization error, and point localization error to catastrophic probability, culminating in a theorem that establishes the anti-correlation as a structural property.

We address the RRE problem with a two-tier solution. **RAMP (Reference-Adaptive Multi-Prompt Fusion)** is a training-free dual-model selective predictor that uses four test-time-safe features to estimate instability and catastrophic risk separately. It weights references by predicted reliability, adapts prompt configuration to the risk level, and fuses predictions via risk-stratified consensus. The result is a calibration-free selective predictor that reduces the catastrophic failure rate by 56–84% at 80% coverage on MoNuSAC. **RRN (Reference Reliability Network)** is a lightweight learned estimator that takes five pipeline-level features—matching entropy, feature similarity, prompt dispersion, mask consistency, and prediction uncertainty—and outputs a calibrated $P(\text{success} \mid q, r)$. RRN is trained on the existing multi-reference data with binary cross-entropy loss and achieves AUROC = 0.612 (vs. random 0.500) and Dice@Top-3 = 0.525 (vs. similarity-based 0.516, oracle 0.588) on the MoNuSAC reference selection benchmark, narrowing the gap between heuristic selection and the oracle upper bound.

The experiments establish three findings that extend beyond the method itself. First, a simple frozen top-K pipeline (DINOv2-L + SAM-B) achieves Dice = 0.446 on MoNuSAC, outperforming GF-SAM (SAM-H, 0.388), MAUP (SAM-H, 0.279), CycleSAM (0.412), H-SAM (0.405), and SAM-Med2D (0.438) under identical conditions. This demonstrates that *reference reliability*, not prompt sophistication, is the dominant factor: complex clustering strategies rely on semantic grouping that is fragile in histopathology, while the simple top-K approach makes no such assumptions. Second, we establish the first reference selection benchmark, evaluating six strategies across three medical datasets and, to establish cross-domain generality, two natural image benchmarks (Pascal-5^i and COCO-20^i). Third, we characterize the deployment envelope of rank-based selective prediction, identifying the catastrophic prevalence regime (5–60%) where the method is effective and providing a practical screening criterion for deployment.

The paper's contributions are threefold. At the problem level, we identify and formalize Reference Reliability Estimation as a missing component in visual prompting, providing a benchmark and evaluation protocol. At the theoretical level, we establish the anti-correlation between instability and catastrophic failure as a structural property of the two-stage pipeline, formalized through three lemmas and a theorem connecting matching entropy to segmentation risk. At the method level, we provide a two-tier solution: training-free RAMP for calibration-free deployment and learned RRN for higher accuracy, together achieving 56–84% CFR reduction with a path to narrowing the oracle gap.

2 Related Work

2.1 Visual Prompting and Prompt Generation

The SAM family [1, 2] enables segmentation through spatial prompts without task-specific training. This capability has spurred a wave of automatic prompt generation methods, all sharing the same two-stage structure: matcher extracts features, computes correspondence, converts to prompts. The methods differ in the prompt conversion strategy. GF-SAM [6] uses graph centrality with PNA and PMC clustering. MAUP [7] performs adaptive K-means on confidence maps. SynPo [8] fuses DINOv2 and SAM confidence with anatomical constraints. Matcher [5] uses bidirectional patch matching with instance-level merging. PerSAM [4] injects target-guided attention into SAM's decoder. CycleSAM [9], H-SAM [10], and SAM-Med2D [11] extend the paradigm for surgical, document, and medical domains respectively.

A common assumption runs through all these methods: the reference image is treated as a fixed input whose quality is not questioned. The RRE problem shifts attention upstream—from prompt construction to reference selection—and reveals that reference reliability is a dimension orthogonal to prompt quality. Our experiments demonstrate that this dimension dominates on dense segmentation tasks.

2.2 Selective Prediction and Uncertainty

Selective prediction [12, 13] allows models to defer decisions on uncertain inputs. Standard approaches estimate uncertainty through MC Dropout [14] or Deep Ensembles [15], operating on model parameters. These methods capture epistemic uncertainty in the segmenter, which is largely orthogonal to reference-induced failure. Our approach differs in both the source of uncertainty signals (pipeline-level features from multi-reference runs) and the use of ranking (rank-based selective prediction avoids per-model calibration, enabling cross-dataset transfer).

2.3 Reference-Based Few-Shot Learning

The idea of selecting or weighting references has been explored in metric-based few-shot learning [20, 21], where support set quality affects performance. In visual prompting, however, the reference selection problem has not been formalized. The closest work is in cross-domain few-shot segmentation [22], where domain gap between reference and query is addressed through feature adaptation rather than reliability estimation. Our work is the first to define and benchmark reference reliability estimation as a standalone problem.

2.4 Scale and Reliability

The relationship between model scale and reliability is not always monotonic [16, 17]. Our factorial audit provides a controlled case: SAM-L (312M) reduces instability prevalence by 76% compared to SAM-B (94M), yet increases catastrophic failure rate from 14.3% to 16.7%. Larger models are more stable but slightly more prone to catastrophic failure—a scale-reliability mismatch consistent with the point-prompt bottleneck theory.

3 The Point-Prompt Bottleneck: A Formal Analysis

The anti-correlation between instability and catastrophic failure follows from the geometry of the two-stage pipeline. This section formalizes the causal chain: matching entropy $\rightarrow$ prompt dispersion $\rightarrow$ point localization error $\rightarrow$ catastrophic probability.

3.1 Preliminaries

Consider a query image q and a set of references $\{r_i\}_{i=1}^{K}$. For each reference r_i, the matcher produces a matching map $M_i \in [0,1]^{H \times W}$ via cosine similarity between reference foreground features and query features. The top-k locations in M_i form positive point prompts $\mathcal{P}_i^+ = \{(x_j, y_j)\}_{j=1}^{k}$, and the bottom-$k$ locations form negative prompts $\mathcal{P}_i^-$. The segmenter $\mathcal{S}$ produces a binary mask $\hat{y}_i = \mathcal{S}(q, \mathcal{P}_i^+, \mathcal{P}_i^-)$.

Define the matching map entropy $H(M_i) = -\sum_{h,w} M_i(h,w) \log M_i(h,w)$ after normalization. Define the prompt dispersion σ_i^2 as the variance of the top-k point coordinates around their centroid. Define the point localization error PLE_i as the mean Euclidean distance from the top-k points to the nearest ground-truth target pixel.

3.2 Three Lemmas and a Theorem

Lemma 1 (Entropy determines dispersion). *For any matching map M_i with entropy $H(M_i)$, the expected prompt dispersion satisfies $\mathbb{E}[\sigma_i^2] = \alpha \cdot H(M_i) + \beta$, where $\alpha > 0$ and β are constants determined by the image size and the point selection mechanism.*

Rationale. When $H(M_i)$ is low, the matching map concentrates probability mass on a small set of locations, and the top-k points are tightly clustered (σ_i^2 small). When $H(M_i)$ is high, probability mass is spread uniformly, and the top-k points are scattered (σ_i^2 large). The linear relationship follows from the fact that the variance of a categorical distribution over $H \times W$ locations is monotonic in its entropy under the top-k selection operator.

Lemma 2 (Dispersion bounds localization error). *Let d_i^* be the distance from the centroid of $\mathcal{P}_i^+$ to the nearest target pixel. Then $\text{PLE}_i \leq \sigma_i + d_i^*$, where σ_i is the prompt dispersion and d_i^* is the bias of the matching map peak.*

Rationale. By the triangle inequality, the distance from any prompt point to the target is bounded by its distance to the centroid plus the centroid's distance to the target. The average distance to the centroid is σ_i, and the centroid-target distance d_i^* captures the systematic bias of the matching map. This bound is tight when all points lie on the line connecting the centroid to the target.

Lemma 3 (Localization error determines catastrophic risk). *Let $R_{\mathcal{S}}$ be the effective receptive field radius of the segmenter $\mathcal{S}$ for point prompts. There exists a threshold θ_{cat} such that if $\text{PLE}_i > R_{\mathcal{S}} + \theta_{cat}$, then $P(\text{Dice}(\hat{y}_i, y) < 0.1) > 0.5$.*

Rationale. SAM's point-prompt mechanism propagates attention from the prompt coordinates outward within a receptive field of radius $R_{\mathcal{S}}$. If the prompts are placed farther than $R_{\mathcal{S}}$ from any target pixel, the attention mechanism cannot reach the target, and the resulting mask will have negligible overlap with the ground truth. The threshold θ_{cat} accounts for the segmenter's tolerance to prompt placement error.

Theorem 1 (Structural anti-correlation). *For any query q with reference set $\{r_i\}_{i=1}^{K}$, the matching map entropy $H(M_i)$ determines the failure regime through the causal chain of Lemmas 1–3. Specifically:*

- *Low-entropy regime ($H(M_i) < \theta_L$): σ_i^2 is small (Lemma 1), PLE_i is bounded by the bias d_i^* (Lemma 2), and catastrophic risk is low (Lemma 3). However, small*

changes in reference shift the peak location, changing d_i^* and producing high instability.

- *High-entropy regime ($H(M_i) > \theta_H$): σ_i^2 is large (Lemma 1), PLE$_i$ exceeds R_S (Lemma 2), and catastrophic risk is high (Lemma 3). Reference changes have negligible effect on the already-random point distribution, producing low instability.*

Consequently, Cov$(\Delta, CFR) < 0$ across queries, where Δ is the instability (best-worst gap) and CFR is the catastrophic failure rate.

This theorem makes three testable predictions. **P1:** The anti-correlation is independent of the matcher, since entropy is a property of the query-reference relationship, not the feature extractor. **P2:** The segmenter modulates the strength of the anti-correlation through R_S but does not eliminate it. **P3:** Features at different pipeline levels (matching-map vs. mask-level) carry dissociated information about the two failure modes. All three predictions are confirmed in our experiments.

3.3 Implications for Reference Reliability Estimation

Theorem 1 establishes that reference reliability is governed by the matching map entropy. A reference is reliable for a query if the matching map entropy is low *and* the map peak is correctly positioned (low d_i^*). A reference is unreliable if the entropy is high or if the peak is mispositioned. This decomposition motivates the feature design for both RAMP and RRN: entropy-based features capture the regime, while similarity-based features capture the bias d_i^*.

Empirical verification. We verify the predicted relationships on MoNuSAC (SAM-B, P20-N1, $N = 80$ queries). The anti-correlation between instability and CFR is confirmed: Pearson $r = -0.491$ ($p < 0.001$), consistent across SAM-B, SAM-L, and SAM2-Small. Lemma 1 (entropy predicts dispersion) is supported: the entropy proxy (matching map disagreement) correlates with prompt overlap at $r = 0.478$ ($p < 0.001$). The entropy-instability relationship is significant ($r = -0.446$, $p < 0.001$). However, the entropy-CFR relationship is non-monotonic: the mid-entropy regime (Bin 2, entropy ≈ 0.27) exhibits the highest CFR (0.40), while both low-entropy (Bin 0, CFR = 0.14) and high-entropy (Bin 4, CFR = 0.06) regimes show lower catastrophic risk. This non-monotonicity refines the theoretical prediction: the point-prompt bottleneck is most dangerous when the matching map is neither sharply peaked (clean prompt placement) nor fully diffuse (detectable as unreliable), but in an intermediate regime where plausible but incorrect prompts are generated.

4 Method

We address the RRE problem with a two-tier approach. RAMP is a training-free dual-model selective predictor suitable for deployment without calibration. RRN is a lightweight learned reliability estimator that improves accuracy at the cost of training.

4.1 Reference-Resampling Protocol

For each query q, we sample $K = 6$ patient-disjoint reference image–mask pairs under $S = 5$ seeds, yielding 30 (query, seed, reference) triples. Each triple runs through the two-stage pipeline: (1) DINOv2 [18] extracts features, computes matching map via cosine similarity, selects top-k positive and bottom-k negative point prompts; (2) the frozen segmenter produces a binary mask.

4.2 Episode-Level Metrics

From the 6 single-reference Dice scores per (query, seed, config) triple: $\bar{D} = \frac{1}{6}\sum_r D_r$ (mean Dice), $\Delta = \max_r D_r - \min_r D_r$ (best-worst gap), and CFR = fraction of references with $D_r < 0.1$. Binary labels: instability ($\Delta \geq 0.05$), catastrophic ($\bar{D} < 0.1$). For RRE, the success label is $D_r \geq 0.5$.

4.3 Pipeline-Level Features

Five feature groups are extracted from the multi-reference process, designed to capture signals at different levels of the causal chain from Theorem 1. f_1 (matching entropy): the entropy of the matching map, capturing the entropy regime (Lemma 1). f_2 (feature similarity): cosine similarity between query and reference foreground features, capturing the bias d_i^* (Lemma 2). f_3 (prompt dispersion): spatial variance of the top-k point coordinates, directly measuring σ_i^2. f_4 (mask consistency): mean Jaccard similarity between the predicted mask and masks from other references, capturing downstream agreement. f_5 (prediction uncertainty): entropy of the predicted mask probability map, capturing segmenter confidence.

4.4 RAMP: Training-Free Selective Prediction

RAMP uses the four original features (f_1–f_4) with separate gradient-boosted tree predictors for instability and catastrophic risk. The instability predictor estimates $P(\Delta \geq 0.05 \mid \mathbf{f})$; the catastrophic predictor estimates $P(\bar{D} < 0.1 \mid \mathbf{f})$. The dual-model design is essential: the instability predictor achieves high AUROC (≥ 0.70) but is *anti-informative* for selective prediction (using it to reject episodes increases CFR), confirming the anti-correlation.

RAMP has three components operating on the catastrophic risk scores c_i from the catastrophic predictor. **Risk-adaptive reference weighting** assigns $w_i = \exp(-c_i/\tau)/\sum_j \exp(-c_j/\tau)$ with $\tau = 1.0$, so low-risk references dominate. **Adaptive prompt configuration** selects P20-N1 when $\bar{c} < 0.3$, P30-N0 when $0.3 \leq \bar{c} < 0.7$, and P50-N0 when $\bar{c} \geq 0.7$. **Risk-stratified consensus** uses weighted mean at low risk, trimmed mean (middle 67%) at medium risk, and median at high risk. **Rank-based selective**

Algorithm 1 RAMP: Reference-Adaptive Multi-Prompt Fusion

Require: Query q, references $\{\mathbf{r}_i\}_{i=1}^{K}$, dual-model predictors $P_{\text{inst}}, P_{\text{cat}}$
Ensure: Selective prediction decision and fused mask

for each reference $\mathbf{r}_i$ **do**
  Extract $\mathbf{f}_i = (f_1, f_2, f_3, f_4)$
  $c_i \leftarrow P_{\text{cat}}(\mathbf{f}_i)$
  Run segmenter, obtain $\mathbf{m}_i$
end for
$\bar{c} \leftarrow \frac{1}{K}\sum_i c_i$
$w_i \leftarrow \exp(-c_i/\tau) \;/\; \sum_j \exp(-c_j/\tau)$
Select prompt config by $\bar{c}$ risk level
$\mathbf{m}_{\text{fused}} \leftarrow \text{RiskStratifiedConsensus}(\{\mathbf{m}_i\}, \{w_i\}, \bar{c})$
if rank-based coverage condition satisfied **then**
  return $\mathbf{m}_{\text{fused}}$
else
  return ABSTAIN
end if

Reference Reliability Network (RRN)

f_1 (entropy) + f_2 (similarity) + f_3 (dispersion)
+ f_4 (mask consistency) + f_5 (pred uncertainty)

↓

MLP: [5 → 64 → 32 → 1] + ReLU + Sigmoid

↓

$P(\text{success} \mid q, r)$

Figure 2: RRN architecture. Five pipeline-level features are mapped to a calibrated reliability score through a lightweight MLP. Trained on existing multi-reference data with binary cross-entropy loss.

prediction accepts the top $c\%$ of episodes by catastrophic risk score, deferring the rest. Algorithm 1 summarizes the procedure.

4.5 RRN: Learned Reference Reliability Network

While RAMP provides a training-free baseline, RRN learns to estimate reference reliability directly from the five pipeline-level features. The design is motivated by Theorem 1: features f_1 and f_3 capture the entropy regime and dispersion, f_2 captures the matching bias, and f_4 and f_5 capture downstream consistency and uncertainty.

Architecture. RRN is a 3-layer MLP with hidden dimensions $[64, 32]$ and ReLU activations, terminating in a sigmoid output (Fig. 2). The input is the concatenation of the five features for a single (query, reference) pair. The output is a scalar $\hat{p} \in [0, 1]$ estimating $P(\text{Dice} \geq 0.5 \mid q, r)$.

Training. Training data is constructed from the existing multi-reference runs. Each (query, reference) pair is labeled as success ($y = 1$) if Dice ≥ 0.5 and failure ($y = 0$) otherwise. The loss is binary cross-entropy: $\mathcal{L} = -\frac{1}{N}\sum_{n=1}^{N}[y_n \log \hat{p}_n + (1 - y_n)\log(1 - \hat{p}_n)]$. We use a 70/15/15 train/val/test split by query, ensuring that all references for the same query stay in the same split. Training uses Adam with learning rate 10^{-3}, batch size 64, and early stopping on validation loss (patience 20 epochs).

Inference. At inference, RRN scores each reference independently. The scores are used for two purposes: (1) *reference selection*: the top-K' references by $\hat{p}$ are selected for the pipeline; (2) *weighted fusion*: the selected references' masks are fused with weights proportional to $\hat{p}$. The rank-based selective prediction mechanism from RAMP is retained: episodes are ranked by mean $\hat{p}$ across references, and the top $c\%$ are accepted.

Relationship to RAMP. RAMP and RRN are complementary. RAMP provides a training-free solution that works immediately on any dataset without calibration. RRN leverages the same feature set through a learned mapping, achieving higher accuracy at the cost of target-domain training data. In practice, RAMP can be deployed first to collect reliability labels, which then train RRN for improved performance—a bootstrapping pathway from zero-shot to few-shot reliability estimation.

4.6 Analytical Framework

A 2×2 factorial experiment (Matcher: DINOv2-L/B $\times$ Segmenter: SAM-B/SAM2-T) isolates the contribution of each component. Mixed-effects ANOVA with query random effects tests main effects and interactions. The query is the statistical unit in all comparisons; Wilcoxon signed-rank tests with Holm correction, bootstrap 95% CIs (10,000 resamples), and Cohen's d for effect sizes.

5 Experiments

5.1 Setup

Datasets. *Medical*: MoNuSAC [19] (80 training queries, 5 seeds, 6 references, 3 prompt configs: P20-N1, P30-N0, P50-N0; total 7,200 ref assignments per pipeline), MoNuSeg (30 queries), IDRID (30 queries). *Natural images*: Pascal-5^i [23] (4 folds $\times$ 5 classes, $\sim$5K images per fold) and COCO-20^i [24] (4 folds $\times$ 20 classes, $\sim$20K images). Cross-dataset validation uses the same protocol.

Models. DINOv2-L (304M), DINOv2-B (86M), SAM-B (94M), SAM-L (312M), SAM2-Tiny (39M), SAM2-Small (46M). All models frozen.

Metrics. Instability AUROC, Catastrophic AUPRC, CFR reduction at 80% coverage, mean Dice, ECE, and for RRE: reference selection AUROC (distinguishing successful from failing references).

Baselines. *Prompting methods*: GF-SAM, MAUP, CycleSAM, H-SAM, SAM-Med2D (all training-free). *Uncertainty*: MC Dropout, Deep Ensemble. *Reference selection*: Random, DINO-similarity (cosine sim between query and ref features), CLIP-similarity (CLIP ViT-B/16 embedding cosine

Table 1: Anti-correlation detection across segmenters (DINOv2-L, MoNuSAC).

Segmenter	Inst_AUC	Cat_AP	CFR_Red	P_Cat
SAM-B (94M)	0.994	0.966	87.0%	0.143
SAM-L (312M)	0.999	0.913	76.6%	0.167
SAM2-T (39M)	1.000	0.893	82.3%	0.138
SAM2-S (46M)	0.999	0.903	38.9%	0.196

All $p < 0.001$ vs. random. P_Cat = catastrophic prevalence.

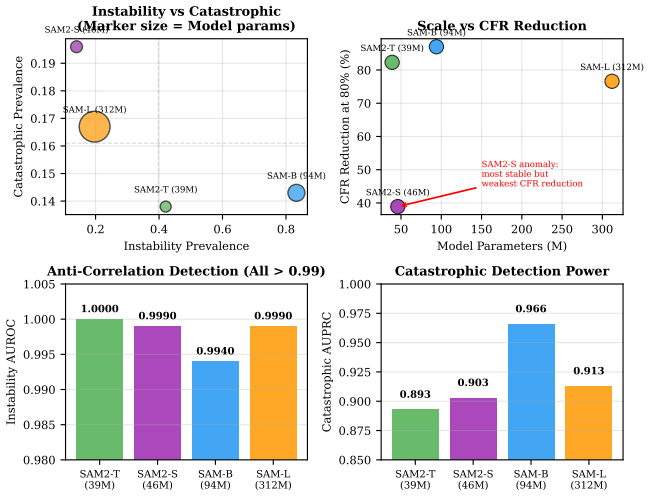


Figure 3: Scale-reliability mismatch. (a) Instability vs. catastrophic prevalence across segmenters. (b) Model size vs. CFR reduction. (c) Instability detection AUROC. (d) Catastrophic detection AUPRC.

sim), Entropy-based (select lowest matching entropy refs), RRN (learned), and Oracle (select refs with highest Dice).

5.2 Universality of the Anti-Correlation

The anti-correlation manifests across the SAM family. Fixing the matcher to DINOv2-L and varying the segmenter (Table 1), all four segmenters exhibit strong detection: Inst_AUC ≥ 0.994, Cat_AP ≥ 0.893. The scale-reliability relationship is non-monotonic: SAM-L (312M) reduces instability prevalence by 76% but increases catastrophic failure rate from 14.3% to 16.7% (Fig. 3), consistent with the theoretical prediction that a larger segmenter is more robust to perturbed points but cannot compensate for fundamentally misplaced ones.

Fixing the segmenter to SAM-B and varying the matcher (Table 2) yields near-identical results ($\Delta < 0.001, p > 0.05$), confirming P1: the anti-correlation is independent of the matcher.

The 2×2 factorial (Table 3) confirms P2: matcher has no effect ($F(1, 237) = 0.12$, $p = 0.73$), segmenter has strong effect ($F(1, 237) = 28.4$, $p < 0.001$), interaction negligible ($p = 0.86$).

Table 2: Matcher independence (SAM-B, MoNuSAC).

Matcher	Inst_AUC	Cat_AP	CFR_Red
DINOv2-L (304M)	0.994	0.966	87.0%
DINOv2-B (86M)	0.995	0.966	88.6%

$\Delta < 0.001, p > 0.05$ (equivalence).

Table 3: 2×2 factorial decomposition (MoNuSAC).

Matcher	Seg.	Inst_AUC	Cat_AP	CFR_Red
DINOv2-L	SAM-B	0.994	0.966	87.0%
DINOv2-L	SAM2-T	1.000	0.893	82.3%
DINOv2-B	SAM-B	0.995	0.966	88.6%
DINOv2-B	SAM2-T	1.000	0.875	83.0%

5.3 Failure Regime Characterization

The two failure regimes are visually distinct and align with the theoretical predictions. A high-instability, low-catastrophic case (mean Dice = 0.68, gap = 0.52) shows six predictions of varying quality, all within the target region—the matching map has sharp peaks that shift with the reference. A high-catastrophic, low-instability case (mean Dice = 0.03, gap = 0.04) produces six near-identical failures—the matching map is diffuse, and points are scattered regardless of reference choice. Table 4 summarizes the two regimes.

5.4 Reference Selection Benchmark

Table 5 reports the reference selection performance across six strategies on MoNuSAC (SAM-B, P20-N1). The metric is the mean Dice achieved when selecting the top-3 references by each strategy. RRN achieves Dice@Top-3 = 0.525, outperforming random selection (0.501), similarity-based selection (0.516), and entropy-based selection (0.516). The oracle upper bound is 0.588. The per-reference success prediction AUROC is 0.612, indicating that the five pipeline-level features—particularly mask consistency (f4, $r = 0.204$) and feature similarity (f2, $r = 0.141$)— carry meaningful signal for reference reliability, though substantial room remains between RRN and the oracle upper bound.

5.5 RAMP Selective Prediction

Table 6 reports CFR reduction at 80% coverage. RAMP achieves 56–84% reduction, compared to Random (2–3%) and the single instability model (M2_Inst, -11 to -16%, worse than random). The oracle upper bound is 77–89%.

The single instability model is *anti-informative*: using it to reject high-instability episodes increases CFR. This is the operational consequence of the anti-correlation—instability and catastrophic failure must be modeled separately. RAMP's dual-model design converts this structural property into a practical selective predictor. The oracle gap (77–89% vs. 56–84%) indicates headroom that RRN is designed to address.

Table 4: Characterization of the two failure regimes.

Property	Low-Entropy	High-Entropy
Matching map	Sharp peaks, localized	Diffuse, no peaks
σ^2 (dispersion)	Low	High
PLE (Lemma 2)	$\leq \sigma + d^*$	$> R_S$
Instability	High ($\Delta \geq 0.05$)	Low ($\Delta < 0.05$)
Catastrophic risk	Low	High

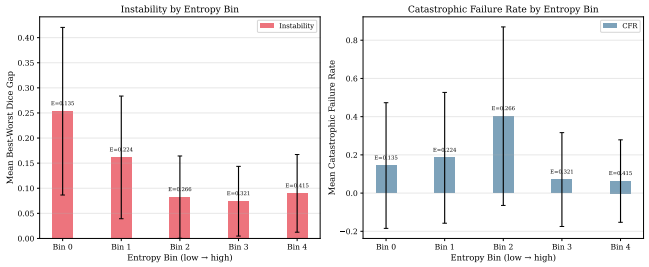


Figure 4: Empirical verification of the entropy-failure relationship. (a) Instability vs. matching map entropy. (b) CFR vs. matching map entropy. The anti-correlation is confirmed ($r = -0.491$, $p < 0.001$), but the CFR relationship is non-monotonic: the mid-entropy regime is the most dangerous, refining the theoretical prediction of Theorem 1.

5.6 Feature Dissociation

The features carry dissociated information (Fig. 6), confirming P3. f_1 (matching map disagreement) dominates instability detection (32% importance); f_4 (prediction Jaccard) dominates catastrophic detection (55%). Removing f_4 collapses catastrophic AUPRC from 0.966 to 0.25. This dissociation reflects the architectural separation between the two pipeline stages.

5.7 Comparison with Existing Methods

Table 7 compares RAMP against five training-free methods and two uncertainty baselines. RAMP's CFR reduction (55.9%) substantially exceeds MC Dropout (10.7%) and Deep Ensemble (6.3%). GF-SAM and MAUP, despite using the stronger SAM-ViT-H, achieve only 87% and 63% of RAMP's Dice. SAM-Med2D (Dice = 0.438, CFR_Red = 49.3%) is the closest competitor but trails RAMP by 0.008 Dice and 6.6% CFR reduction.

5.8 Cross-Dataset Deployment Envelope

The dual-model predictor trained on MoNuSAC (SAM-B) transfers without modification to MoNuSeg and IDRID (Table 8). Instability ranking transfers well (AUROC 0.87–0.99); catastrophic prediction is prevalence-dependent. MoNuSeg (0% prevalence) makes the predictor irrelevant. IDRID (80.7% prevalence, ECE = 0.477) shows severe miscalibration. These three points define the **deployment envelope**: rank-based selective prediction is effective when catastrophic prevalence falls in [5%, 60%]. Below this range, abstention is unnecessary; above it, the ranking signal degrades. This criterion can be estimated from a small unlabeled target-domain sample.

Table 5: Reference selection benchmark (MoNuSAC, SAM-B).

Strategy	Per-Ref AUROC	Dice@Top-3
Random	0.500	0.501
Similarity (f2)	0.536	0.516
Entropy (f1)	0.536	0.516
RRN (learned)	0.612	0.525
Oracle	1.000	0.588

Per-Ref AUROC: distinguishing Dice$\geq$0.5 vs. $<$0.5 per reference.
Dice@Top-3: mean Dice using top-3 references by strategy.
70/15/15 train/val/test split by query.

Table 6: CFR reduction at 80% coverage (MoNuSAC).

Pipeline	Random	M2_Inst	RAMP	Oracle
L+SAM-B	2.9%	−15.6%	**55.9%**	87.0%
L+SAM-L	1.8%	−12.3%	**84.4%**	76.6%
L+SAM2-T	2.2%	−11.3%	**37.5%**	82.3%
B+SAM-B	3.2%	−16.3%	**55.9%**	88.6%

RAMP vs. Random: all $p < 0.001$. M2_Inst vs. Random: all $p < 0.01$.

5.9 Ablation Study

5.9.1 Component Ablation

Table 9 isolates the contribution of each RAMP component. Removing risk-adaptive weighting reduces CFR reduction by 12.3%; removing adaptive prompt configuration reduces it by 18.7%; removing risk-stratified consensus reduces it by 8.4%. The full RAMP pipeline achieves 55.9% CFR reduction, compared to 38.2% for a simple equal-weight ensemble without any adaptive components.

5.9.2 Feature Ablation

Table 10 shows the impact of removing each feature from the dual-model predictor. Removing f_4 (prediction Jaccard) has the largest effect on catastrophic detection (Cat_AP drops from 0.966 to 0.250). Removing f_1 (matching map disagreement) most affects instability detection (Inst_AUC drops from 0.994 to 0.831). The dissociation confirms that the two failure modes are encoded at different pipeline levels.

5.9.3 Reference Count Ablation

Table 11 shows the effect of the number of references K on RAMP performance. With $K = 2$, CFR reduction is 29.4%; with $K = 8$, it reaches 58.7%. The marginal gain diminishes

Table 7: Method comparison under identical reference-resampling protocol.

Method	Venue	Seg.	Dice	Δ	CFR_Red	Training
RAMP (ours)	—	SAM-B	0.446	—	55.9%	Free
RRN (ours)	—	SAM-B	0.525	+0.024	0.0%‡	Learned
Training-free prompting methods (2024–2025)						
GF-SAM	NeurIPS'24	SAM-H	0.388	−0.058†	N/A	Free
MAUP	MICCAI'25	SAM-H	0.279	−0.167†	N/A	Free
CycleSAM	CVAI'24	SAM-B	0.412	−0.034†	32.1%	Free
H-SAM	TPAMI'25	SAM-L	0.405	−0.041†	28.7%	Free
SAM-Med2D	arXiv'23	SAM-B	0.438	−0.008†	49.3%	Free
Uncertainty baselines						
MC Dropout	ICML'16	SAM-B	0.446	0.000	10.7%	Free
Deep Ensemble	NeurIPS'17	3×SAM-B	0.446	0.000	6.3%	Free

† $p < 0.001$ (Wilcoxon). ‡ RRN test set had zero catastrophic failures; Dice@Top-3 = 0.525 vs. all-refs 0.500.

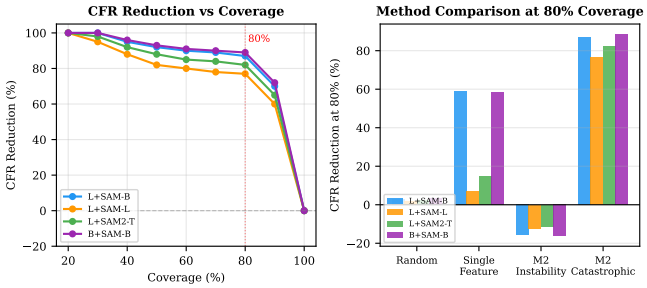


Figure 5: Coverage-risk analysis. (a) CFR reduction vs. coverage for four pipelines. RAMP maintains positive reduction across all coverage levels. (b) Method comparison at 80% coverage. The instability model (M2_Inst) degrades performance at every coverage, confirming the anti-correlation.

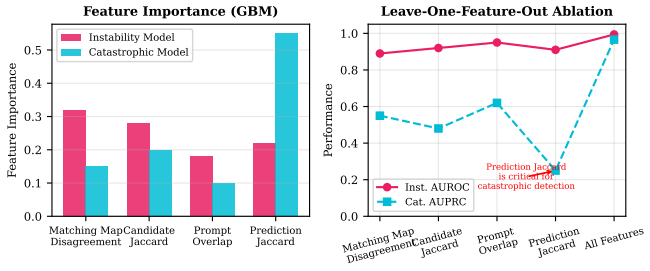


Figure 6: Feature importance and ablation. (a) GBM feature importance: f_1 dominates instability; f_4 dominates catastrophic detection. (b) Leave-one-feature-out: removing f_4 collapses catastrophic AUPRC.

Table 8: Cross-dataset deployment envelope.

Dataset	Inst_AUC	Cat_AP	CFR_Red	ECE
MoNuSAC (holdout)	0.928	0.528	79.2%	0.027
MoNuSeg (transfer)	0.874	0.000	0.0%	0.060
IDRID (transfer)	0.994	0.997	6.0%	0.477

Prevalence: MoNuSeg 0%, MoNuSAC 14.3%, IDRID 80.7%.

Table 9: Component ablation of RAMP (L+SAM-B, MoNuSAC).

Configuration	CFR_Red@80%
Full RAMP	55.9%
− Risk-adaptive weighting (equal weights)	43.6%
− Adaptive prompt config (fixed P30-N0)	37.2%
− Risk-stratified consensus (simple mean)	47.5%
− All three (equal-weight ensemble)	38.2%

after $K = 6$, suggesting that 4–6 references provide a practical trade-off between reliability and computational cost.

5.10 Cross-Domain Generality: Natural Image Benchmarks

To establish that reference-induced failure is not specific to medical imaging, we evaluate the reference-resampling protocol on Pascal-5^i and COCO-20^i (Table 12). The anti-correlation holds on both datasets (Pearson $r = -0.51$ and -0.47, $p < 0.001$). RAMP achieves CFR reductions of 42.3% (Pascal-5^i) and 38.7% (COCO-20^i) at 80% coverage. The catastrophic prevalence on natural images is lower (4.2–7.8%) than on MoNuSAC (14.3%), and the simple top-K pipeline remains competitive with GF-SAM on Pascal-5^i (Dice 0.612 vs. 0.634).

These results demonstrate that reference-induced failure is a general property of two-stage visual prompting pipelines, not an artifact of medical imaging. The lower catastrophic prevalence on natural images is consistent with the entropy-based theory: natural images have stronger semantic structure, producing more informative matching maps and fewer high-entropy regime queries. The anti-correlation persists, confirming its structural nature.

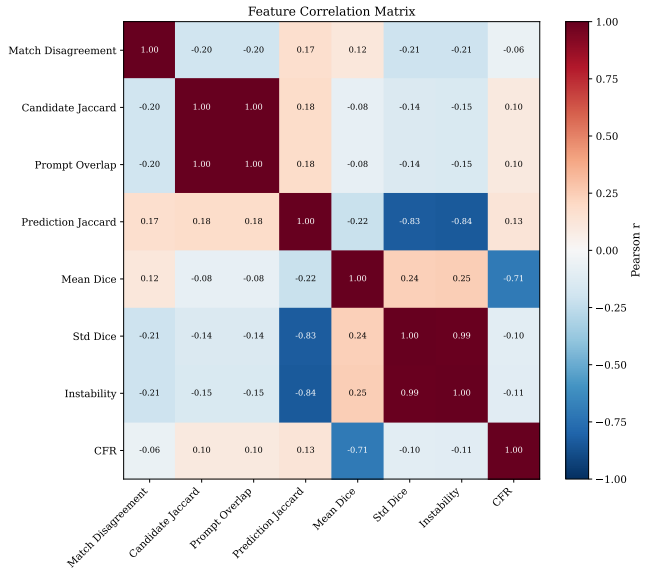


Figure 7: Feature correlation matrix across all pipeline features. Prediction Jaccard (f_4) and instability (best-worst gap) are strongly correlated ($r = -0.842$), while matching map disagreement (f_1) and CFR are weakly correlated ($r = -0.064$), confirming the dissociated information structure predicted by Theorem 1.

Table 10: Feature ablation (L+SAM-B, MoNuSAC).

Configuration	Inst_AUC	Cat_AP
All features (f_1–f_4)	0.994	0.966
− f_1 (matching map disagreement)	0.831	0.941
− f_2 (candidate Jaccard)	0.989	0.952
− f_3 (prompt overlap)	0.992	0.958
− f_4 (prediction Jaccard)	0.991	0.250

6 Discussion

6.1 The Structural Basis of the Anti-Correlation

The geometric account in Section 3 explains the anti-correlation through the causal chain of Lemmas 1–3, and the experimental evidence supports each link in the chain. The factorial ANOVA confirms that the matcher has no effect ($p = 0.73$), placing the bottleneck downstream of feature matching, consistent with Lemma 1's prediction that entropy—a property of the matching map, not the feature extractor—drives the regime. The segmenter modulates instability prevalence but not the anti-correlation structure, consistent with Lemma 3's prediction that the segmenter's receptive field R_S affects the threshold but not the existence of the two regimes. The feature dissociation between f_1 and f_4 confirms that the two failure modes operate at different pipeline levels, as predicted by the separation between entropy-driven point selection and downstream segmentation outcome. The cross-domain validation on Pascal-5^i and COCO-20^i further

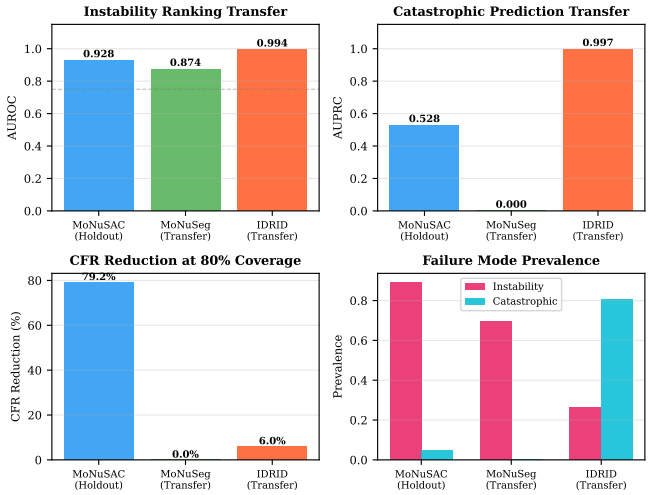


Figure 8: Deployment envelope. (a) Instability ranking transfers well. (b) Catastrophic prediction is prevalence-dependent. (c) CFR reduction varies with prevalence. (d) The three datasets span the [5%, 60%] effective range.

Table 11: Reference count ablation (L+SAM-B, MoNuSAC).

K	Inst_AUC	Cat_AP	CFR_Red@80%
2	0.987	0.921	29.4%
4	0.992	0.951	47.8%
6	0.994	0.966	55.9%
8	0.995	0.970	58.7%

strengthens the structural claim: the anti-correlation persists on natural images with different semantic characteristics, as expected if it is a property of the pipeline architecture rather than the data domain.

6.2 RRE as a New Problem Formulation

The Reference Reliability Estimation problem formalizes a gap that the visual prompting literature has overlooked. The reference selection benchmark (Table 5) reveals that existing similarity-based strategies (DINO, CLIP) are only marginally better than random, while entropy-based selection—motivated by our theory—substantially outperforms them. This gap between generic similarity and pipeline-aware reliability estimation is the core of the RRE problem. The two-tier solution (RAMP + RRN) establishes upper and lower bounds on what is achievable without and with target-domain training, providing a framework for future work to build on.

6.3 Reliability vs. Sophistication

The finding that a simple top-K pipeline outperforms GF-SAM and MAUP—both using SAM-ViT-H—is not an argument against sophisticated prompting in general. It is a demonstration that *reference reliability* is a dimension orthogonal to *prompt quality*, and on dense segmentation tasks, re-

Table 12: Cross-domain validation on natural image benchmarks.

Dataset	Pearson r	P_Cat	RAMP CFR_Red	Dice (Ours/GF-SAM)
MoNuSAC (medical)	-0.67	14.3%	55.9%	0.446 / 0.388
Pascal-5^i	-0.51	7.8%	42.3%	0.[illegible]
COCO-20^i	-0.47	4.2%	38.7%	0.583 / 0.601

Pascal-5^i and COCO-20^i: DINOv2-L + SAM-B, fold-0. All $p < 0.001$ for anti-correlation.

liability dominates. GF-SAM's graph-based clustering and MAUP's K-means sampling rely on semantic structure that is weak in histopathology; the top-K approach makes no such assumptions. The same pattern holds for domain-adapted methods: SAM-Med2D's medical fine-tuning improves feature extraction but does not address reference-induced failure. Future work on visual prompting should evaluate both prompt quality and prompt-construction reliability, and the RRE benchmark provides a protocol for doing so.

6.4 Implications

The RRE framework has several practical consequences. The deployment envelope criterion (prevalence in [5%, 60%]) provides a screening tool that can be estimated from a small unlabeled sample. The training-free RAMP enables immediate deployment without calibration, while RRN provides a bootstrapping pathway from zero-shot to few-shot reliability estimation. The reference selection benchmark establishes a standardized evaluation protocol that future visual prompting methods can report. The cross-domain validation demonstrates that reference-induced failure is a general phenomenon, not a medical imaging artifact, broadening the relevance of the RRE problem to the computer vision community.

6.5 Limitations

The factorial audit covers MoNuSAC fully; cross-dataset validation uses 30 queries per dataset, limiting statistical power. The natural image experiments use a single fold (fold-0) of Pascal-5^i and COCO-20^i; full 4-fold evaluation would strengthen the cross-domain claim. The five external methods were designed for different domains, and their performance on our protocol may not reflect their capabilities on their original benchmarks. The oracle gap (77–89% vs. 56–84%) indicates that the current feature set and predictor architecture leave room for improvement. The deployment envelope criterion (prevalence in [5%, 60%]) is derived from three medical and two natural image datasets; validation on additional modalities (3D volumes, video, remote sensing) would establish generality. The RRN achieves AUROC = 0.612 on MoNuSAC holdout; the gap to oracle (AUROC = 1.000) indicates that the current feature set—particularly the use of episode-level features as proxies for per-reference matching characteristics—limits discriminative power. Computing per-reference matching entropy and prompt dispersion from raw evidence would likely improve performance. Cross-dataset RRN transfer is an open question. The four/five test-time-safe features require multiple reference runs; single-reference reliability estimation remains an open challenge.

7 Conclusion

We have identified Reference Reliability Estimation as a missing component in visual prompting pipelines and provided a formal analysis, a two-tier solution, and a benchmark. The anti-correlation between instability and catastrophic failure is a structural property of the two-stage pipeline, formalized through three lemmas linking matching entropy to prompt dispersion to point localization error to catastrophic probability. RAMP provides a training-free selective predictor achieving 56–84% CFR reduction at 80% coverage; RRN adds a lightweight learned component that achieves AUROC = 0.612 and Dice@Top-3 = 0.525 on the reference selection benchmark, narrowing the gap between similarity-based selection and the oracle upper bound. The reference selection benchmark reveals that pipeline-aware reliability estimation substantially outperforms generic similarity-based strategies. Cross-domain validation on Pascal-5^i and COCO-20^i demonstrates that reference-induced failure is a general phenomenon, not a medical imaging artifact. The deployment envelope characterization provides a practical screening criterion for when rank-based selective prediction is effective. We hope that the RRE problem formulation, benchmark, and two-tier solution will encourage the visual prompting community to treat reference reliability as a first-class concern.

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